

Verifying an agentic GIS routing workflow

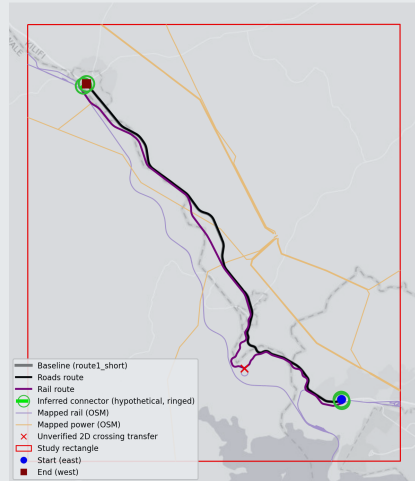
Fiber route comparison on the shortened Mombasa corridor: baseline, roads, power and rail

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GGs 662: Agentic GeoAI · Assignment 2

September 28, 2026

Week 4 audit: shortened corridor routes, connectors and flagged crossings



Route and Infrastructure data © OpenStreetMap contributors (ODbL). Basemap: Esri World Light Gray Canvas (Esri, HERE, Garmin, © OpenStreetMap contributors).

Map data © OpenStreetMap contributors (ODbL); basemap Esri

Agent framework

Goal

Compare baseline, roads, power and rail routes between the exact route1_short endpoints (east = start); report length and USD 10/m cost.

Agency

Agent cleans, nodes, repairs gaps and routes. **Human** fixed endpoints, 50 m gap limit, 500 m endpoint limit and the rate.

Environment

Saved o3_01 OpenStreetMap (OSM) GeoPackages, AGENTS.md rules, 3,000 m study rectangle, measured in EPSG:32737. No new downloads.

Feedback

Stop with endpoint_too_far, no_path or empty_network. A failed check means revise, never relax a threshold.

Tools

geopandas, shapely, networkx (Dijkstra), pandas; matplotlib + contextily for maps.

Verification

Independent audit script, synthetic cases, controlled reruns. Real-world claims need imagery or authoritative data.

Where the human decides and where the agent revises

Stays with a human

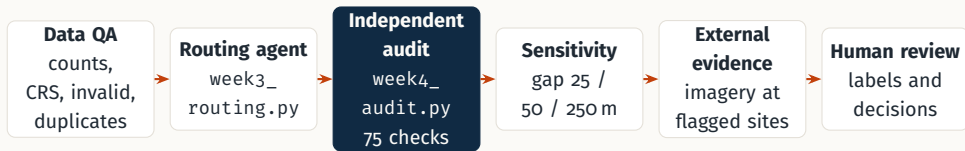
Whether an inferred connector or a 2D crossing is actually buildable, and which corridor to pursue.

Feedback that changes the next action

The audit flags the rail route switching lines at a bridge ($-3.99374, 39.54626$) with no shared OSM node.

Imagery showed an overpass, so the agent should **remove that transfer and reroute rail**, then report the new length or `no_path`. It should not raise a threshold to keep the old answer.

Approach: checks designed around the agent, not after it

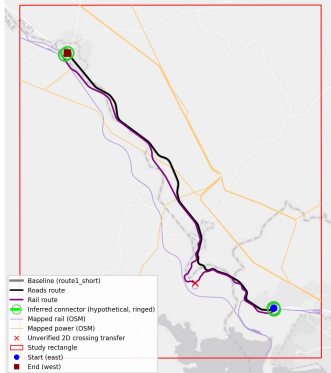


Independent of the code. The audit never imports the routing script; it recomputes every value from the saved files and the 03_01 inputs.

Not independent of the author. The same AI tool wrote both scripts, so they can share a wrong assumption, such as treating every 2D crossing as a junction. The audit can only confirm the computations, so the flagged sites were also checked in satellite imagery, which is independent of both OSM and the code.

Route map: what the agent built and what it inferred

Week 4 audit: shortened corridor routes, connectors and flagged crossings



Route and infrastructure data © OpenStreetMap contributors (ODbL). Basemap: Esri World Light Gray Canvas (Esri, HERE, Garmin, © OpenStreetMap contributors).

Green rings: inferred connectors. Roads 0.59 m and 0.72 m; rail 194.8 m and 372.7 m. All four are endpoint connectors: hypothetical new construction.

Red cross: flagged crossing. The rail route switches onto a bridge-tagged line where the two lines share no OSM vertex.

Lengths. Baseline 22,896.8 m; roads 22,897.6 m; rail 25,902.4 m. Power: endpoint_too_far.

Data © OpenStreetMap contributors (ODbL).
Basemap: Esri World Light Gray Canvas.

Verification evidence: 75 reproducible checks

73

PASS

2

FAIL: rail crossing (audit +
imagery)

0

NOT_CHECKED

Category	PASS	FAIL	NOT_CHK	What it checks
Provenance and data QA	7	0	0	counts, CRS, invalid geometry, duplicates
Comparison table and cost	9	0	0	four rows, blank unavailable values, length × USD 10/m
Route geometry and length	23	0	0	endpoints, continuity, metres, edge sum within 0.01 m
Connectors	6	0	0	limits, containment, counted once
Unavailable route (power)	1	0	0	endpoint distance recomputed
Synthetic cases	20	0	0	6 cases × 3 runs, plus the 230 m oracle
Sensitivity	5	0	0	only the gap setting changed
Spatial assumption	0	1	0	route transfers only at OSM junctions
External evidence	2	1	0	satellite imagery at the 3 flagged sites

What the checks caught

FAIL: a 2D crossing at a rail bridge

The route moves from way 891041250 onto bridge-tagged way 239559205 at (−3.993745, 39.54626); neither line has a vertex there. Imagery shows an **overpass**: the route cannot switch lines here.

A bug in the agent's own checker

The connector-limit check hard-coded 50 m, so the 250 m rerun was checked against the wrong limit. It now reads the run's setting.

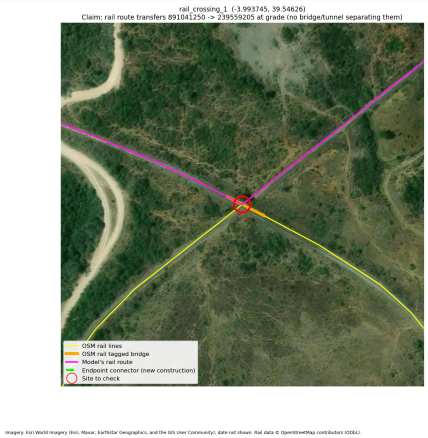
A wrong claim in the Week 3 answers

The write-up said rail used a gap connector. `connectors.gpkg` shows all four used connectors are endpoint connectors (rail: 194.8 m and 372.7 m). Corrected.

A justified failure: power

`endpoint_too_far` confirmed independently: start 1,839.2 m and end 995.1 m from the nearest mapped power line, against a 500 m limit.

External evidence: the flagged crossing is an overpass



Claim tested in imagery	Result
Rail route switches lines at grade at the flagged crossing	FAIL overpass
No rail closer than the 195 m start connector	PASS
No rail closer than the 373 m end connector	PASS

Source: Google Maps satellite (Maxar, Airbus; ©2026), checked 27 Sep 2026; independent of OSM.
Picture: Esri imagery with the model's lines.

Both OSM ways are the metre-gauge main line, and Google labels the spot “Mazeras Railway Spiral”: the line loops over itself. The audit’s FAIL was a real error.

Sensitivity: changing only the gap tolerance changes no route outcome

Gap tolerance	Roads	Power	Rail	Power parts*	Ranking
25 m	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	6	baseline < roads < rail
50 m (original)	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	3	baseline < roads < rail
250 m	ok, 22,897.6 m	endpoint_too_far	ok, 25,902.4 m	1	baseline < roads < rail

*Power connected components after gap repair (9 before). Runs: sensitivity_gap_25m, default, gap_tolerance_250m.

Why nothing moves. No route uses a gap connector, and power fails on the 1,839 m endpoint distance, which no gap setting can fix. Connector counts and lengths are identical in all three runs.

What this does not show. One setting at three values is not global robustness (Saltelli et al., 2019). A common rate of USD 5, 10 or 20/m scales costs but keeps the same order.

Conclusions: what I trust and what stays unresolved

Conclusion	Label (o4_o2 §16)	Evidence
Lengths, endpoints, containment and costs are computed correctly	VERIFIED	R-*, C-*
Connectors within limits, counted once; all are endpoint connectors	VERIFIED	R-*-conn*
Power unavailable because its start is 1,839 m from mapped lines	VERIFIED ; a real line there: UNRESOLVED	U-power
Ranking baseline < roads < rail holds for a 25–250 m gap tolerance	not SENSITIVE to this one setting	S-*
The rail route is continuous through the crossing	CONTRADICTED : imagery shows an overpass	V-crossings, E-crossing
Rail endpoint connectors are new construction, not unmapped track	EXTERNALLY SUPPORTED (one source, one date)	E-connector 3, 4
Access rights, permits, feasibility and real costs	UNRESOLVED	none in data

So the rail route as computed is **not valid**: the agent should forbid that transfer and re-route (or report no_path), not relax a threshold. Roads stays the route to investigate.

Sources and AI use

Data, tools and readings

- ▶ Roads, rail, power, settlements: © OpenStreetMap contributors, ODbL (saved in o3_o1).
- ▶ Basemaps: Esri World Light Gray Canvas and World Street Map (Esri, HERE, Garmin, © OpenStreetMap contributors).
- ▶ GGS 662 notebooks o3_o1, o3_o2, o4_o1, o4_o2 and AGENTS.md. Python: geopandas, shapely, networkx, pandas, pyproj.
- ▶ Oreskes, Shrader-Frechette & Belitz (1994), *Science*, doi:10.1126/science.263.5147.641
- ▶ Haklay (2010), *Environment and Planning B*, doi:10.1068/b35097
- ▶ Saltelli et al. (2019), *Environmental Modelling & Software*, doi:10.1016/j.envsoft.2019.01.012

AI-use statement

Tool: Claude Code (Anthropic, Claude Opus 5.5).

How it contributed: general coding assistance, writing and debugging the Week 3 routing script, the Week 4 audit, and the plotting and export scripts, and running the reruns; drafting the expected-checks table; and generating these slides.

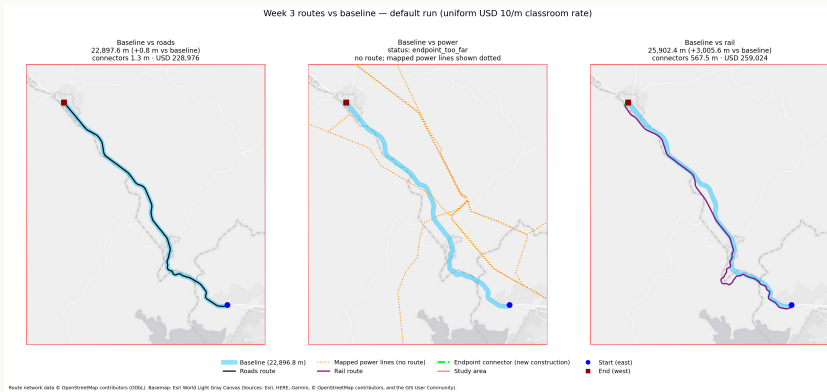
My role: I directed the analysis, checked the three flagged sites in satellite imagery myself, reviewed the results and directed edits to the slides. I remain responsible for understanding, testing and validating all submitted material. Because the same tool wrote the routing code and the audit, I treat the audit as verification, not independent validation.

Thank you

Questions?

Supporting slides follow: full check table, synthetic cases and evidence files

Supporting: each candidate route against the baseline



Roads follows the baseline almost exactly (+0.8 m). Power has no route: its start is 1,839 m from the nearest mapped line. Rail is 3,006 m longer, and imagery shows it is **not followable** at the Mazeras crossing. Source: route_side_by_side.png, 50 m run.

Supporting: expected checks, written as specifications (04_02 §13)

Check	Expected / tolerance	Failure would mean	Audit IDs
Endpoints	Easterly start, westerly end within 0.01 m	Wrong problem solved	R*-ends
Continuity, validity	One valid LineString, no gaps or loops	Length and cost meaningless	R*-valid, -simple
Containment	Inside baseline bounds + 3,000 m	Data outside allowed environment	R*-contain
Metres	EPSG:32737; edge sum within 0.01 m; UTM vs geodesic $<0.5\%$; $\geq$ straight line	CRS or double-count error	R*-len, -units
Connectors	Gap $\leq$ run tolerance; endpoint $\leq$ 500 m; counted once; baseline 0	Invented or double-counted construction	R*-conn*, C-base
Cost	length $\times$ USD 10 within USD 0.01, one rate	Wrong ranking	R*-cost, C-rate
Unavailable	Named status, blank length/cost, no geometry	Failure hidden as a route	C-unavail, U-power
Synthetic	5 AGENTS.md cases; 80+120+30 m $\rightarrow$ 230 m, USD 2,300	Logic bug hidden by real data	T-*
Sensitivity	Only gap tolerance changes (25 m); changes reported	Uncontrolled or forced result	S-*
Junctions	Transfers only at shared OSM vertices	Bridge or tunnel treated as a junction	V-crossings

Geometry cannot establish: buildability, access rights or permits; a 2D crossing may be a bridge (needs imagery, E-*). Full table: `Claude/04_Project/expected_checks.ipynb`.

Supporting: per-route checks (all PASS)

Check (tolerance)	Baseline	Roads	Rail
Valid, nonempty single LineString	PASS	PASS	PASS
Endpoints match baseline (≤ 0.01 m)	0.0000 / 0.0000 m	0.0000 / 0.0000 m	0.0000 / 0.0000 m
No loops or self-crossing (<code>is_simple</code>)	True	True	True
Inside study rectangle	True	True	True
Geometry vs traversed-edge sum (≤ 0.01 m)	0.000000 m	0.009833 m	0.007000 m
Metres not degrees (UTM vs geodesic, $< 0.1\%$)	-0.036%	-0.036%	-0.036%
Not shorter than straight line	1.111×	1.111×	1.257×
Cost = length $\times$ USD 10/m	228,967.56	228,975.93	259,023.67
Connectors counted once (count, total)	0, 0 m	2, 1.3139 m	2, 567.5396 m
Each connector ≤ 500 m, inside, on route, at endpoint	-	0.59, 0.72 m	194.85, 372.69 m
Non-connector length on mapped lines (± 1 m)	-	22,896.4 m	25,334.9 m

Source: `validation_table.csv` (R-* checks). Note: roads edge-sum difference (0.0098 m) passes but sits close to the 0.01 m tolerance, consistent with 1 cm node rounding.

Supporting: provenance, data QA and comparison table

Check ID	Observed	Status
P-settings	Settings match spec: gap 50 m, endpoint 500 m, margin 3,000 m, EPSG:32737	PASS
D-roads	4,742 features, EPSG:4326, 0 missing / empty / invalid, all intersect study area	PASS
D-power	15 features, 0 missing / empty / invalid	PASS
D-rail	77 features, 0 missing / empty / invalid	PASS
D-baseline	1 LineString, valid	PASS
D-settlements	30 features (29 points, 1 polygon), map context only	PASS
D-roads-dup	2,292 of 4,742 edges are reversed OSMnx duplicates; removed before routing	PASS
C-rows	baseline, roads, power, rail: one row each	PASS
C-cols	nine required CSV columns	PASS
C-base	baseline connector values 0, 0, 0	PASS
C-unavail	power: endpoint_too_far, blank length and cost (not zero)	PASS
C-gpkg	routes.gpkg holds baseline, roads, rail	PASS

Supporting: unavailable route, spatial assumption, sensitivity, external evidence

Check ID	Observed	Status
C-rate	one common rate, USD 10/m	PASS
U-power	start 1,839.2 m, end 995.1 m from nearest power line (>500 m)	PASS
V-crossings	1 transfer at a non-shared vertex; involves a bridge-tagged way	FAIL
S-25m-ctrl	only "Gap connector limit" changed, 50 → 25 m	PASS
S-25m-out	no differences; ranking unchanged; power parts 6	PASS
S-250m-ctrl	only "Gap connector limit" changed, 50 → 250 m	PASS
S-250m-out	no differences; ranking unchanged; power parts 1	PASS
S-rate	same order at USD 5, 10, 20/m	PASS
E-rail connector 3	195 m connector (−4.008849, 39.591099): no track closer than the mapped line	PASS
E-rail connector 4	373 m connector (−3.864413, 39.472359): mapped line is the closest rail	PASS
E-rail crossing 1	crossing (−3.993745, 39.54626): overpass, not a junction	FAIL

Source: Google Maps satellite imagery (Maxar, Airbus; ©2026, credit year, not exact capture date), independent of OSM. Recorded in `external_evidence.csv`; site maps `site_*.png`.

Supporting: synthetic cases and known-answer oracle

Case (AGENTS.md)	Expected	50 m run	25 m run	250 m run
Already connected: components	1	1 PASS	1 PASS	1 PASS
Gap below threshold: components after repair	1	1 PASS	1 PASS	1 PASS
Gap above threshold: components after repair	2	2 PASS	2 PASS	2 PASS
Endpoint on a line interior: status	ok	ok PASS	ok PASS	ok PASS
Disconnected endpoints: status	no_path	no_path PASS	no_path PASS	no_path PASS
Connector counted once: route length (10 m + gap + 20 m)	= expected	55.0 m PASS	42.5 m PASS	155.0 m PASS
Oracle: 80 m + 120 m + 30 m gap at USD 10/m	230 m, USD 2,300	230 m, USD 2,300 PASS		
Oracle fault: gap double-counted	flagged	260 m $\neq$ 230 m flagged PASS		

The gap in the connector case is half the run's tolerance, so the expected length differs per run. Source: `routing_report.md` in each run folder; audit IDs T-*

Supporting: evidence files, run commands and versions

Run commands (every argument has a default)

- ▶ `python Claude/03_Project/week3_routing.py → outputs/default/`
- ▶ `python week3_routing.py --gap-tolerance-m 25 --out outputs/sensitivity_gap_25m`
- ▶ `python week3_routing.py --gap-tolerance-m 250 --out outputs/gap_tolerance_250m`
- ▶ `python Claude/04_Project/week4_audit.py → outputs/week4_validation/`

Evidence files

- ▶ `outputs/week3_routing/` and each rerun subfolder: `route_comparison.csv`, `routes.gpkg`, `connectors.gpkg`, `route_map.png`, `routing_report.md`
- ▶ `outputs/week4_validation/`: `week4_audit.py`, `validation_table.csv`, `validation_report.md`, `validation_map.png`, `inferred_crossings.gpkg`, `external_evidence.csv`

Versions: Python 3.12.2; geopandas 1.1.4; shapely 2.1.2; pyproj 3.8.0; networkx 3.6.1; pandas 2.3.3.

Output location: as O4_O2 specifies, under the repo root: `outputs/week3_routing/` (original run, plus `sensitivity_gap_25m/` and `gap_tolerance_250m/`) and `outputs/week4_validation/`. Copied from the working folders under Claude/ by Claude/04_Project/export_evidence.py.